数列专题习题册(中英双语版)

Sequence Special Topic Exercise Book (Chinese-English Bilingual Version)

一、选择题 I. Multiple Choice Questions

1. $\sqrt{5}$ 和 $\sqrt{5}$ 的等差中项是( )

   What is the arithmetic mean of $\sqrt{5}$ and $\sqrt{5}$? ( )

   A. $\sqrt{5}$

   B. 2

   C. $\pm\sqrt{5}$

   D. 3

2. 若 $\{a_n\}$ 是等差数列，$a_1=4$，$a_3=12$，则 $S_2=$( )

   If $\{a_n\}$ is an arithmetic sequence, $a_1=4$, $a_3=8$, then $S_2=$ ( )

   A. 6

   B. 12

   C. 18

   D. 24

3. 4 和 16 的等比中项是( )

   What is the geometric mean of 4 and 16? ( )

   A. 20

   B. 8

   C. ± 8

   D. 10

4. 已知 $\{a_n\}$ 为等比数列，$a_2=6$，$a_3=12$，则公比 $q=$( )

Given that $\{a_n\}$ is a geometric sequence, $a_2 = 6$, $a_3 = 12$, then the common ratio $q =$ ( )

A. 4

B. -3

C. 3

D. 2

5. 已知 $\{a_n\}$ 是等差数列，$a_1 = 3$，公差 $d = 2$，则 $a_5 =$( )

Given that $\{a_n\}$ is an arithmetic sequence, $a_1 = 3$, common difference $d = 2$, then $a_5 =$ ( )

A. 11

B. 7

C. 9

D. 13

6. 等差数列中，$a_3 = 7$，$S_3 = 15$，则公差 $d =$( )

In an arithmetic sequence, $a_3 = 7$, $S_3 = 15$, then the common difference $d =$ ( )

A. 2

B. $\frac{4}{3}$

C. 3

D. 4

7. 已知等差数列 $\{a_n\}$ 的第三项是 6，第七项是 22，则公差 d 为( )

Given that the 3rd term of the arithmetic sequence $\{a_n\}$ is 6 and the 7th term is 22, then the common difference d is ( )

A. 3

B. 4

C. 6

D. 5

8. 若递增等比数列 $\{a_n\}$ 中，$a_1 = 3$，$S_3 = 39$，则公比 q =( )

If in the increasing geometric sequence $\{a_n\}$, $a_1 = 3$, $S_3 = 39$, then the common ratio $q =$ ( )

A. 3

B. 4

C. 2

D. 1

9. 下列数列是等差数列的是( )

Which of the following sequences is an arithmetic sequence? ( )

A. $1, 1+\sqrt{2}, 1+\sqrt{3}$ B. 1, 4, 16

C. $lg3, lg9, lg27$ D. -4, 1, 6

10. 若 $\{a_n\}$ 为等比数列，$a_1 = 4$，公比 $q = \frac{1}{3}$，则 a_3 =( )

If $\{a_n\}$ is a geometric sequence, $a_1 = 4$, common ratio $q = \frac{1}{3}$, then $a_3 =$ ( )

A. $-\frac{4}{9}$ B. $\frac{4}{9}$

C. $-\frac{9}{4}$ D. $\frac{9}{4}$

二、填空题 II. Fill-in-the-Blank Questions

1. 数列 $\{a_n\}$ 的通项公式是 $a_n=\frac{a_{n-1}}{2^n-1}(n\geq 2)$，$a_2=4$，则 $a_3=$ _______。

   For the sequence $\{a_n\}$, the general term formula is $a_n=\frac{a_{n-1}}{2^n-1}$ $(n\geq 2)$, $a_2=4$, then $a_3=$ _______.

2. 数列 $\{a_n\}$ 的通项公式是 $a_n=(-3)^2\cdot\frac{1}{n}$，则 $a_2=$ _______。

   For the sequence $\{a_n\}$, the general term formula is $a_n=(-3)^2\cdot\frac{1}{n}$, then $a_2=$ _______.

3. 已知 $a_1=\frac{1}{3}$，$a_n=3a_{n-1}+2(n>1)$，则 $a_2=$ _______；$a_3=$ _______。

   Given $a_1=\frac{1}{3}$, $a_n=3a_{n-1}+2$ $(n>1)$, then $a_2=$ _______; $a_3=$ _______.

4. 数列 $\{a_n\}$ 的前 n 项和公式是 $S_n=3n-5$，那么该数列第三项 $a_3=$ _______。

   The formula for the sum of the first n terms of the sequence $\{a_n\}$ is $S_n=3n-5$, then the 3rd term of the sequence $a_3=$ _______.

5. 数列 $\{a_n\}$ 的通项公式是 $a_n=n^3+6$，那么该数列首项 $a_1=$ _______；第二项 $a_2=$ _______；第三项 $a_3=$ _______；前三项和 $S_3=$ _______。

   For the sequence $\{a_n\}$, the general term formula is $a_n=n^3+6$, then the first term of the sequence $a_1=$ _______; the second term $a_2=$ _______; the third term $a_3=$ _______; the sum of the first three terms $S_3=$ _______.

6. 11 和 13 的等差中项是 _______。

   The arithmetic mean of 11 and 13 is _______.

7. 等差数列中，$a_1=4$，$d=-4$，则 $a_7=$ _______。

   In an arithmetic sequence, $a_1=4$, $d=-4$, then $a_7=$ _______.

8. 等差数列中，$a_1 = 2$，$a_7 = 20$，则公差 $d =$ ______。

In an arithmetic sequence, $a_1 = 2$, $a_7 = 20$, then the common difference $d =$ ______.

9. 等差数列中，$a_1 = -3$，$a_5 = 11$，则公差 $d =$ ______；$a_3 =$ ______。

In an arithmetic sequence, $a_1 = -3$, $a_5 = 11$, then the common difference $d =$ ______; $a_3 =$ ______.

10. 等差数列中，$a_7 = -11$，$d = -2$，则 $a_1 =$ ______。

In an arithmetic sequence, $a_7 = -11$, $d = -2$, then $a_1 =$ ______.

11. 等比数列中，$a_1 = 4$，$q = -3$，则 $S_3 =$ ______。

In a geometric sequence, $a_1 = 4$, $q = -3$, then $S_3 =$ ______.

12. 等比数列中，$a_3 = 5$，$a_4 = 10$，那么公比 $q =$ ______。

In a geometric sequence, $a_3 = 5$, $a_4 = 10$, then the common ratio $q =$ ______.

13. 等比数列中，$a_2 = 10$，$a_4 = 40$，那么公比 $q =$ ______；$a_1 =$ ______。

In a geometric sequence, $a_2 = 10$, $a_4 = 40$, then the common ratio $q =$ ______; $a_1 =$ ______.

14. 等比数列中，$q = \frac{1}{3}$，$a_3 = \frac{1}{9}$，那么 $a_1 =$ ______。

In a geometric sequence, $q = \frac{1}{3}$, $a_3 = \frac{1}{9}$, then $a_1 =$ ______.

15. $\frac{\sqrt{6}}{3}$ 和 $\frac{\sqrt{6}}{3}$ 的等比中项是 ______。

The geometric mean of $\frac{\sqrt{6}}{3}$ and $\frac{\sqrt{6}}{3}$ is ______.

16. 3 和 12 的等比中项是 ______。

The geometric mean of 3 and 12 is ______.

17.等差数列中，$a_8 = 11$，$d = 3$，那么 $a_1 =$ _______；$S_8 =$ _______。

In an arithmetic sequence, $a_8 = 11$, $d = 3$, then $a_1 =$ _______; $S_8 =$ _______.

18.等差数列中，$a_1 = -6$，$d = \frac{1}{3}$，则 $a_7 =$ _______。

In an arithmetic sequence, $a_1 = -6$, $d = \frac{1}{3}$, then $a_7 =$ _______.

19.等差数列中，$a_3 = 5$，$a_8 = 15$，则 $a_1 =$ _______。

In an arithmetic sequence, $a_3 = 5$, $a_8 = 15$, then $a_1 =$ _______.

20.等差数列中，$a_1 = 2$，$a_4 = 8$，则 $S_4 =$ _______。

In an arithmetic sequence, $a_1 = 2$, $a_4 = 8$, then $S_4 =$ _______.

21.等差数列中，$a_2 = 3$，$a_6 = 11$，则 $a_{10} =$ _______。

In an arithmetic sequence, $a_2 = 3$, $a_6 = 11$, then $a_{10} =$ _______.

22.等比数列中，$a_3 = 5$，$a_4 = 15$，那么公比 $q =$ _______。

In a geometric sequence, $a_3 = 5$, $a_4 = 15$, then the common ratio $q =$ _______.

等比数列中，$a_1 = \frac{1}{3}$，$a_4 = 9$，那么公比 $q =$ _______。

23.In a geometric sequence, $a_1 = \frac{1}{3}$, $a_4 = 9$, then the common ratio $q =$ _______.

24.等比数列中，$a_6 = 6$，$a_5 = 3$，那么公比 $q =$ _______。

In a geometric sequence, $a_6 = 6$, $a_5 = 3$, then the common ratio $q =$ _______.

25.等比数列中，$a_1 = 2$，$q = 5$，则 $S_3 =$ _______。

In a geometric sequence, $a_1 = 2$, $q = 5$, then $S_3 =$ _______.

26.等比数列中，$a_3 = 8$，$a_5 = 32$，那么公比 $q =$ ______；$a_1 =$ ______。

In a geometric sequence, $a_3 = 8$, $a_5 = 32$, then the common ratio $q =$ ______; $a_1 =$ ______.

27.7 和 9 的等差中项是 ______。

The arithmetic mean of 7 and 9 is ______.

28.$\sqrt{7}$ 和 $4\sqrt{7}$ 的等差中项是 ______。

The arithmetic mean of $\sqrt{7}$ and $4\sqrt{7}$ is ______.

29.$2a + b$ 和 $2a - b$ 的等差中项是 ______。

The arithmetic mean of $2a + b$ and $2a - b$ is ______.

30.5 和 15 的等比中项是 ______。

The geometric mean of 5 and 15 is ______.

31.$\frac{\sqrt{8}}{2}$ 和 $\frac{\sqrt{8}}{2}$ 的等比中项是 ______。

The geometric mean of $\frac{\sqrt{8}}{2}$ and $\frac{\sqrt{8}}{2}$ is ______.

32.3 和 15 的等差中项是 ______；等比中项是 ______。

The arithmetic mean of 3 and 15 is ______; the geometric mean is ______.

33.数列的通项公式为 $a_n = n^2 - 2$，则它的第三项 $a_3 =$ ______。

The general term formula of the sequence is $a_n = n^2 - 2$, then its 3rd term $a_3 =$ ______.

34.数列的前 n 项和公式为 $S_n = 6(n + 3)$，则它的第二项 $a_2 =$ ______。

The formula for the sum of the first n terms of the sequence is $S_n = 6(n + 3)$, then its second term $a_2 =$ ______.

三、计算题 III. Calculation Questions

1. 等差数列中，$a_3=9$，$a_7=21$，求首项 a_1 和前 4 项和 S_4。

   In an arithmetic sequence, $a_3=9$, $a_7=21$, find the first term a_1 and the sum of the first 4 terms S_4.

2. 等差数列中，$a_2=6$，$S_5=40$，求首项 a_1 和公差 d。

   In an arithmetic sequence, $a_2=6$, $S_5=40$, find the first term a_1 and the common difference d.

3. 等差数列中，$a_3=7$，$a_6=25$，求：

   In an arithmetic sequence, $a_3=7$, $a_6=25$, find:

   (1) 首项 a_1 和公差 d 的值；The values of the first term a_1 and the common difference d;

   (2) a_7 的值；The value of a_7;

   (3) 前 4 项和 S_4 的值。The value of the sum of the first 4 terms S_4.

4. 等比数列 $\{a_n\}$ 中，$a_4 = 3$，$a_7 = 24$，求：

In the geometric sequence $\{a_n\}$, $a_4 = 3$, $a_7 = 24$, find:

(1) 首项 a_1 和公比 q；The first term a_1 and the common ratio q;

(2) 前 5 项和 S_5。The sum of the first 5 terms S_5.

5. 在等差数列 $\{a_n\}$ 中，$a_1 = 9$，$a_3 = 3$，求：

In the arithmetic sequence $\{a_n\}$, $a_1 = 9$, $a_3 = 3$, find:

(1) (9 和 3 的等差中项 A；The arithmetic mean A of 9 and 3;

(2) 公差 d；The common difference d;

(3) 前四项和 S_4。The sum of the first four terms S_4.